

Clinical Implications of Ultrasound-Based Morphology in Choroidal Melanoma

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Objective: To describe the frequency of different B-scan morphologies and their association with clinical features and outcomes.

Design: Cohort study of patients enrolled in the Prospective Ocular Tumor Study from January 2000 to January 2024 initially seen at Mayo Clinic Rochester.

Participants: Consecutive inclusion of patients with posterior uveal melanoma.

Methods: B-scan ultrasounds were performed by an experienced technician and treatment modalities were implemented by the attending oncologist.

Main Outcome Measures: Tumors were classified by shape as observed on B-scan. Enucleation-free survival, metastasis-free survival (MFS), and overall survival rates were analyzed using Cox-regression models and Kaplan-Meier curves.

Results: Among 1021 cases of uveal melanoma, 739 (72.4%) were dome-shaped, 119 (11.7%) were mushroom-shaped, 85 (8.3%) were multilobulated, 77 (7.5%) were minimally elevated, and 1 (0.1%) was diffuse. The median follow-up duration after presentation was 37 months (interquartile range [IQR], 3–324). The macula was more commonly involved in minimally elevated tumors compared with the other groups (63.6% vs. 13.8%, $P < 0.001$). These tumors also exhibited a larger proportion of high internal reflectivity (13% vs. 2.3%, $P < 0.001$). The multilobulated group exhibited a significantly larger diameter at baseline (median, 15 mm; IQR, 6.1–30), whereas the mushroom-shaped group had greater thickness (median, 7.9 mm; IQR, 1.3–17.3) compared with the other groups ($P < 0.001$). Enucleation-free survival at 36 months was lower for mushroom-shaped (60.1%; 95% confidence interval [CI], 47.7–70.3) and multilobulated tumors (71.1%; 95% CI, 55.7–82.7). At 36 months, multilobulated tumors had lower MFS (68.2%; 95% CI, 55–78.2) and overall survival (73.9%; 95% CI, 59.9–83.64). On multivariate analysis adjusted for tumor thickness and diameter, multilobulated melanomas had a higher risk of metastasis (hazard ratio [HR], 2.08; $P = 0.003$) and death (HR, 2.38; $P < 0.001$).

Conclusions: Choroidal melanoma configuration by B-scan can vary from minimally elevated to dome-shaped to mushroom-shaped or multilobulated. Independent of presenting tumor size, multilobulated morphology was identified as a predictor for metastasis and death. Multilobulated melanomas, identified by a readily available tool such as ultrasonography, warrant a vigilant approach and close monitoring due to a potential association with poor prognosis.

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Uveal melanoma (UM) is the most common primary adult intraocular malignancy and the second most frequent melanoma in the body, with an estimated incidence ranging from 1 to 9 per million people per year.¹ It most frequently affects White patients between the ages of 50 and 70 years.¹ The choroid is the site of origin of >90% of all UMs, whereas only 6% begin confined to the ciliary body, and 4% to the iris.¹

Currently, choroidal UM diagnosis is primarily made by clinical examination with assistance of multimodal imaging. Fundoscopic evaluation typically reveals a pigmented, dome-shaped nodular mass under the retinal pigment epithelium with a highly variable degree of pigmentation.² A rupture of Bruch's membrane, suggested by a

mushroom-shaped lesion, is observed in 20% of cases.³ However, tumors often manifest in different shapes or smaller sizes, which can complicate the diagnosis. In such cases, imaging studies render valuable insights. The 10-MHz B-scan ultrasonography probe is an essential evaluation tool that can detect tumors with thickness >1.5 mm.² This technique can reveal different tumor shapes, such as the classically described mushroom or collar-stud appearance, a dome-shaped lesion, or, less commonly, multilobulated or minimally elevated tumors. These less common presentations may not always evoke an immediate clinical concern for melanoma, and in some cases, such as amelanotic multilobulated tumors, can create diagnostic confusion with other diagnoses such as choroidal